QUBO Formulation: Multi-Vehicle Delivery Assignment Problem

Multi-Vehicle Delivery Assignment Problem

Problem Statement. Given a set of vehicles $\mathcal{V}$ and a set of customer jobs $\mathcal{J}$, a collection of candidate routes $\mathcal{R}$ is provided. Each route $r \in \mathcal{R}$ is associated with a delivery cost $c_r \geq 0$ and covers a subset of jobs $J_r \subseteq \mathcal{J}$. Each route is assigned to a specific vehicle $v(r) \in \mathcal{V}$. The objective is to select a subset of routes such that every job in $\mathcal{J}$ is covered exactly once, each vehicle in $\mathcal{V}$ is assigned at most one route, and the total delivery cost is minimized.

Decision Variables. For each candidate route $r \in \mathcal{R}$, introduce a binary decision variable $x_r \in \{0, 1\}$, where $x_r = 1$ indicates that route r is selected, and $x_r = 0$ otherwise.

QUBO Derivation. Step 1 — The original objective is to minimize the total delivery cost:

$$\min_{\mathbf{x} \in \{0,1\}^{|\mathcal{R}|}} \sum_{r \in \mathcal{R}} c_r x_r. \tag{1}$$

Step 2 — The constraints are expressed in general algebraic form as:

$$\sum_{r \in R_j} x_r = 1, \quad \forall j \in \mathcal{J}, \tag{2}$$

$$\sum_{r \in R_v} x_r \leq 1, \quad \forall v \in \mathcal{V}, \tag{3}$$

where $R_j = \{r \in \mathcal{R} : j \in J_r\}$ denotes the set of candidate routes covering job j, and $R_v = \{r \in \mathcal{R} : v(r) = v\}$ denotes the set of candidate routes available for vehicle v.

Step 3 — We convert each constraint into a quadratic penalty term added to the objective. Let $\lambda > 0$ be a penalty weight. For the exact coverage constraints, we use the squared equality penalty:

$$P_j(\mathbf{x}) = \lambda \left(\sum_{r \in R_j} x_r - 1 \right)^2 \tag{4}$$

$$= \lambda \left(\sum_{r \in R_j} x_r^2 - 2 \sum_{r \in R_j} x_r + 1 \right). \tag{5}$$

Using the idempotent property of binary variables, $x_r^2 = x_r$, this expands to:

$$P_j(\mathbf{x}) = \lambda \left(\sum_{r \in R_j} x_r - 2 \sum_{r \in R_j} x_r + 1 \right) \tag{6}$$

$$= \lambda \left(1 - \sum_{r \in R_j} x_r \right) \tag{7}$$

$$= \lambda - \lambda \sum_{r \in R_j} x_r. \tag{8}$$

Note that the squared expansion yields only linear and constant terms; cross-terms $x_r x_{r'}$ cancel out. For the vehicle route limit constraints, we enforce $\sum_{r \in R_v} x_r \leq 1$ by penalizing pairwise selections within the same vehicle:

$$Q_v(\mathbf{x}) = \lambda \sum_{\substack{r, r' \in R_v \\ r < r'}} x_r x_{r'}. \tag{9}$$

This term is zero when at most one route is selected for vehicle v, and strictly positive otherwise.

Step 4 — Combining the objective and all penalty terms yields the single QUBO Hamiltonian:

$$H(\mathbf{x}) = \sum_{r \in \mathcal{R}} c_r x_r + \sum_{j \in \mathcal{J}} \left(\lambda - \lambda \sum_{r \in R_j} x_r \right) + \sum_{v \in \mathcal{V}} \lambda \sum_{\substack{r, r' \in R_v \\ r < r'}} x_r x_{r'}. \tag{10}$$

Step 5 — Collecting terms by variable indices, the QUBO is expressed in the standard form $H(\mathbf{x}) = \sum_r Q_{rr} x_r + \sum_{r<r'} Q_{rr'} x_r x_{r'} + c_0$. The matrix entries and offset are given by:

$$Q_{rr} = c_r - \lambda \cdot |\{j \in \mathcal{J} : r \in R_j\}|, \quad \forall r \in \mathcal{R}, \tag{11}$$

$$Q_{rr'} = \begin{cases} \lambda & \text{if } r, r' \in R_v \text{ for some } v \in \mathcal{V} \text{ and } r < r', \\ 0 & \text{otherwise,} \end{cases} \qquad \forall r, r' \in \mathcal{R}, r < r', \tag{12}$$

$$c_0 = \lambda |\mathcal{J}|. \tag{13}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any constraint violation increases the total energy above the minimum possible objective value. Specifically, let $C_{\max} = \max_{r\in\mathcal{R}} c_r$ be the maximum single-route cost. The maximum possible reduction in the objective function from selecting any subset of routes is bounded by $C_{\max}$. Since each violated constraint contributes at least λ to the energy, choosing $\lambda > C_{\max}$ guarantees that the energy of any infeasible solution strictly exceeds the energy of any feasible solution. Thus, $\lambda > \max_r c_r$ is a sufficient condition.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0,1\}^{|\mathcal{R}|}$ recovers the optimal delivery assignment because (i) for any feasible assignment, all penalty terms vanish, so $H(\mathbf{x})$ equals the total delivery cost; (ii) for any infeasible assignment, at least one constraint is violated, incurring a penalty of at least λ. By the choice of λ, the energy of any infeasible solution is strictly greater than the energy of the best feasible solution. Consequently, the global minimum of $H(\mathbf{x})$ must occur at a feasible point, and among feasible points, $H(\mathbf{x})$ reduces to the original cost function, ensuring the global minimum corresponds to the optimal solution.

Illustrative Example. Consider a small instance with $M = 4$ candidate routes $\mathcal{R} = \{0,1,2,3\}$, costs $c_0 = 2, c_1 = 5, c_2 = 4, c_3 = 1$, two jobs $\mathcal{J} = \{1,2\}$, and two vehicles $\mathcal{V} = \{A,B\}$. The job coverage sets are $R_1 = \{0,1\}$ and $R_2 = \{2,3\}$. The vehicle route sets are $R_A = \{0,2\}$ and $R_B = \{1,3\}$. Substituting these into the general formulas yields:

$$\begin{aligned} H(\mathbf{x}) &= 2x_0 + 5x_1 + 4x_2 + 1x_3 + \lambda(1 - x_0 - x_1) + \lambda(1 - x_2 - x_3) + \lambda x_0 x_2 + \lambda x_1 x_3 \quad (14) \\ &= 2x_0 + 5x_1 + 4x_2 + 1x_3 + 2\lambda - \lambda x_0 - \lambda x_1 - \lambda x_2 - \lambda x_3 + \lambda x_0 x_2 + \lambda x_1 x_3. \quad (15) \end{aligned}$$

The corresponding Q matrix entries and offset are:

$$\begin{aligned} &Q_{00} = 2 - \lambda, \quad Q_{11} = 5 - \lambda, \quad Q_{22} = 4 - \lambda, \quad Q_{33} = 1 - \lambda, \quad (16) \\ &Q_{02} = \lambda, \quad Q_{13} = \lambda, \quad \text{all other } Q_{rr'} = 0, \quad (17) \\ &c_0 = 2\lambda. \quad (18) \end{aligned}$$

Setting $\lambda = 6$ (satisfying $\lambda > \max c_r = 5$), the feasible solutions are $\mathbf{x} \in \{(1,0,0,1), (0,1,1,0)\}$. Evaluating $H(\mathbf{x})$ gives $H(1,0,0,1) = 3$ and $H(0,1,1,0) = 9$. The global minimum is $H = 3$ at $\mathbf{x}^* = (1,0,0,1)$, corresponding to selecting routes 0 and 3 with minimum total cost 3.